

Machine Learning Notes

1. What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, decision-making, language understanding, and perception.

Examples of Artificial Intelligence include virtual assistants, recommendation systems, autonomous vehicles, facial recognition systems, and chatbots.

2. What is Machine Learning?

Machine Learning (ML) is a subset of Artificial Intelligence that allows computers to learn patterns from data without being explicitly programmed.

Instead of writing fixed rules, machine learning algorithms learn from historical data and improve their performance over time.

The three major types of Machine Learning are:

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Applications include spam detection, fraud detection, image recognition, medical diagnosis, and stock market prediction.

3. Supervised Learning

Supervised Learning is a type of machine learning where the model is trained using labeled data.

Examples include:

- House Price Prediction
- Email Spam Detection
- Student Marks Prediction

Popular algorithms include:

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machine (SVM)

4. Unsupervised Learning

Unsupervised Learning works with unlabeled data.

Its objective is to discover hidden patterns or relationships within the data.

Common applications include:

- Customer Segmentation
- Product Recommendation
- Market Basket Analysis

Popular algorithms include:

- K-Means Clustering
- DBSCAN
- Hierarchical Clustering

5. Neural Networks

A Neural Network is a computational model inspired by the human brain.

A neural network consists of:

- Input Layer
- Hidden Layers
- Output Layer

Neural networks are widely used in image recognition, speech recognition, computer vision, and natural language processing.

6. Deep Learning

Deep Learning is a subset of Machine Learning that uses deep neural networks with multiple hidden layers.

Applications include:

- Face Recognition
- Self-Driving Cars
- Medical Imaging
- Language Translation
- Speech Recognition

Popular frameworks include TensorFlow and PyTorch.

7. Gradient Descent

Gradient Descent is an optimization algorithm used to minimize the loss function during training.

It updates the model parameters in the direction that reduces the error.

Update Rule:

$$\text{New Weight} = \text{Old Weight} - \text{Learning Rate} \times \text{Gradient}$$

The learning rate controls the size of each update.

8. Backpropagation

Backpropagation is the algorithm used to train neural networks.

It calculates gradients of the loss function with respect to every weight in the network.

These gradients are then used by Gradient Descent to update the weights.

Without backpropagation, neural networks cannot learn effectively.

9. Overfitting

Overfitting occurs when a model memorizes the training data instead of learning general patterns.

Characteristics include:

- High Training Accuracy
- Low Testing Accuracy

Ways to reduce overfitting include:

- Dropout
 - Regularization
 - Early Stopping
 - Data Augmentation
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10. Evaluation Metrics

Evaluation metrics help measure model performance.

Common metrics include:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

The choice of metric depends on the problem.

11. Natural Language Processing (NLP)

Natural Language Processing (NLP) is a field of Artificial Intelligence that enables computers to understand, analyze, and generate human language.

Applications include:

- Chatbots
- Machine Translation
- Sentiment Analysis
- Text Classification
- Text Summarization

12. Transformers

Transformers are deep learning models introduced in the paper "Attention Is All You Need."

Instead of processing words one by one, transformers use the self-attention mechanism to understand relationships between words.

Popular transformer models include:

- BERT
- GPT
- T5
- LLaMA

Transformers power modern AI systems like ChatGPT and Claude.

13. Large Language Models (LLMs)

Large Language Models are transformer-based neural networks trained on massive datasets.

Examples include:

- GPT
- Gemini
- Claude
- Llama

LLMs can perform:

- Question Answering
- Code Generation
- Text Generation
- Translation
- Summarization

14. Embeddings

Embeddings are numerical vector representations of text.

Instead of storing words directly, embedding models convert sentences into vectors.

Sentences with similar meanings have vectors that are close together in vector space.

Embeddings are the foundation of semantic search and Retrieval-Augmented Generation (RAG).

15. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation combines semantic search with Large Language Models.

Instead of relying only on the model's internal knowledge, RAG retrieves relevant information from external documents before generating an answer.

A typical RAG pipeline consists of:

1. Document Loader
2. Text Splitter
3. Embedding Model
4. Vector Database
5. Retriever
6. Prompt Construction
7. Large Language Model

RAG reduces hallucinations and enables LLMs to answer questions based on private documents.

16. Vector Databases

Vector databases store embeddings instead of traditional rows and columns.

Unlike SQL databases, vector databases perform semantic similarity search.

Popular vector databases include:

- ChromaDB
- FAISS
- Pinecone
- Milvus
- Weaviate

Vector databases are widely used in modern AI applications.

17. ChromaDB

ChromaDB is an open-source vector database designed for AI applications.

It stores:

- Text Chunks
- Embeddings
- Metadata

ChromaDB enables efficient similarity search and integrates seamlessly with LangChain.

18. Similarity Search

Similarity Search compares the embedding of a user's query with the embeddings stored in the vector database.

Instead of searching for exact keywords, similarity search finds documents with similar meanings.

Cosine Similarity is one of the most commonly used techniques for measuring similarity between vectors.

19. LangChain

LangChain is a framework for building applications powered by Large Language Models.

It provides components such as:

- Document Loaders
- Text Splitters
- Prompt Templates
- Retrievers
- Chains
- Memory
- Agents

LangChain simplifies the development of production-ready AI applications.

20. Summary

Artificial Intelligence is the broad field of creating intelligent systems.

Machine Learning enables systems to learn from data.

Deep Learning uses neural networks with many layers.

Transformers power modern language models.

Embeddings convert text into vectors.

Vector Databases store these vectors.

Retrievers find the most relevant information.

Retrieval-Augmented Generation combines retrieval and generation to answer questions using external knowledge.